DATA 557 : Course Project

SEX BIAS ANALYSIS AT A US UNIVERSITY

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1 Background

Disparities in salaries between male and female faculty members at colleges and universities in the United States have been extensively studied. Numerous factors contribute to determining faculty compensation, many of which could overlap with gender-related factors. For instance, salary levels often differ significantly across various disciplines. Typically, faculty members receive salary increments with growing professional experience and upon receiving promotions. This study investigates potential disparities in salary allocation by analyzing salaries at different points in a faculty member's career, including starting salaries, salary growth and promotion rates.

2 Motivation

The issue of salary disparities between male and female faculty members at universities is well-established, prompting ongoing discussions about equity in academia. Although discrimination based on sex is prohibited by law, disparities in faculty salaries often persist due to factors such as differences in experience, degree level, academic discipline, administrative duties, and productivity. This study aims to investigate the presence and extent of sex bias concerning faculty salaries targeting a single U.S. university with particular attention to the following questions:

1. Does sex bias exist in faculty salaries in 1995?
2. Is there evidence of sex bias in initial salaries at the time of hiring?
3. Has sex bias been evident in salary increases between 1990 and 1995?
4. Does sex bias affect the likelihood of promotion from Associate Professor to Full Professor?

3 Dataset Description

The dataset comprises 19,792 records from 1,597 faculty members (excluding medical school faculty) at a U.S. university. The data span from 1976 to 1995 detailing monthly salaries and several potentially influential factors including gender, academic rank, field of expertise, highest degree attained and administrative responsibilities. Missing data is denoted as "NA" which will be addressed in the analysis. The available variables include:

- case: Case number
- id: Unique faculty member identifier
- sex: Gender of faculty (M or F)
- deg: Highest degree attained (PhD, Prof, Other)
- yrdeg: Year when highest degree was attained (2 digits)
- field: Academic field (Arts, Prof, Other)
- startyr: Year faculty member was hired (2 digits)
- year: Year the salary was recorded (2 digits)
- rank: Academic rank (Assistant, Associate, Full)
- admin: Indicator of administrative duties (1 if yes, 0 otherwise)

- salary: Monthly salary in USD

Each row in the dataset represents a faculty member's salary record for a specific year. Multiple records exist per faculty member, allowing for an analysis of salary trends and career progression over time. Time-invariant variables (sex, deg, yrdeg, startyr) remain constant for a faculty member across records. Time-varying variables (rank, admin, salary) change over time, capturing faculty career growth.

Variable	Count	Mean($)	Minimum($)	Maximum($)
Salary	19792	4721.82	1200	14464
Salary (Male)	15866	4854.54	1200	14464
Salary (Female)	3926	4185.46	1267	11036

Table 1: Summary Statistics of Salaries by Gender

4 Methods

The dataset was first cleaned and preprocessed to handle missing values and ensure consistency in variable formats. Year related columns (yrdeg, startyr, year) were standardized by converting them to four digit format. The four records with missing rank values were dropped for consistency. Below are the four questions, each followed by the methods used to address them.

4.1 Does sex bias exist at the university in the most current year available (1995)?

The dataset includes salary information for 1,597 faculty members at a US university in 1995, capturing monthly salaries along with demographic and professional variables such as sex (male/female), rank (Assistant, Associate, Full), administrative duties (yes/no), academic field (Arts, Professional, Other), and highest degree attained (PhD, Professional, Other). The goal was to investigate sex bias in salaries. Preprocessing involved filtering for 1995 records, creating binary variables for categorical predictors (e.g., sex, admin roles), generating dummy variables for multi-category predictors (e.g., rank, degree type, field), handling missing values by excluding incomplete records, and converting all predictors to numeric types for regression analysis.

Descriptive Analyses:

Descriptive analyses were conducted by calculating summary statistics, including mean, median, and standard deviation, separately for each sex. Additionally, several visualizations were generated to provide a clear understanding of the data distribution and patterns. Specifically, box plots were created to illustrate salary distributions by sex and rank, while density plots were used to compare salary distributions between men and women. Bar charts were also developed to visualize gender representation across different ranks and administrative roles.

Inferential Analyses:

Inferential analyses included two-sample t-tests to compare mean salaries between men and women, simple linear regression to quantify unadjusted gender differences, and multiple linear regression models to adjust for confounding factors such as rank, administrative duties, degree type, and academic field. Interaction effects between sex and administrative roles, as well as between sex and academic fields, were explored to identify potential moderating factors. Stratified analyses were conducted within each rank, academic field, and administrative role. Model diagnostics involved residual analysis (e.g., histograms of residuals) to evaluate regression assumptions, assessment of multicollinearity using Variance Inflation Factors (VIF), and application of robust standard errors to address heteroscedasticity.

4.2 Has sex bias existed in the starting salaries of faculty members (salaries in the year hired)?

As an initial preprocessing step, four actions were performed: (1) The dataset was filtered to retain only records with starting salaries; (2) Inflation adjustments were applied to salaries using the CPI module in Python to ensure real purchasing power comparisons over time; (3) An experience variable was created by subtracting the year of degree completion from the start year (startyr - yeardeg). Negative experience values, indicating faculty members completed their degree after being hired, were recoded as a binary variable (1 for degree completed before hire, 0 for after hire); and (4) The start year variable (startyr) was centered by subtracting the mean start year to mitigate multicollinearity for regression analyses.

Descriptive Analysis:

Basic statistical measures including mean, count and standard deviation were calculated separately for male and female faculty. For visual comparisons two types of graphs were used: (1) Scatterplots, plotting start year against salary which allowed for an examination of salary trends over time; (2) Line plots to display the average salary by start year for both genders, providing insight into salary progression before and after adjusting for inflation

Inferential Analysis:

Hypothesis tests and regression analysis was performed to assess the differences in starting salaries between male and female faculty. First, a two-sample t-test for unequal variances (Welch's t-test) was conducted to test the hypothesis of equal mean starting salaries using both raw salary figures and inflation-adjusted salaries. Next, a simple OLS regression was performed to model the effect of sex as a categorical variable on salary. Variance Inflation Factor (VIF) analysis was conducted to ensure that multicollinearity does not compromise the regression results. The start year variable was centered before to reduce its correlation with other predictors. Finally, multiple regression models were applied to control for confounding variables such as field, rank, degree, administrative roles, experience and start year. These models also explored interaction effects to assess the robustness of the observed salary differences.

4.3 Has sex bias existed in granting salary increases between 1990 -1995?

The dataset was filtered for the years 1990 to 1995 to focus on the analysis period. To facilitate salary growth calculations, the dataset was transformed into a wide-format table where each individual had separate salary columns for each year within the timeframe. This structure allowed for the calculation of absolute salary increases, percentage increases relative to the initial salary, and annual salary growth rates (salary slopes).

Descriptive Analysis:

To analyze salary growth patterns summary statistics were computed for salary increases, percentage increases and salary slopes by gender. These included count, mean, standard deviation and percentiles to compare central tendencies and distribution spread. Box plots were used to visualize salary increases, percentage increases, and salary slopes, highlighting differences in distribution and variability. A line plot of average salaries over time provided insights into salary growth trends across years and between groups. Normalized histograms of annual salary growth rates were also generated separately for each group. By accounting for differences in sample size, these density-based visualizations allowed for a clearer comparison of salary growth distributions.

Inferential Analysis:

To assess gender differences in salary growth, a two-sample t-test was conducted to compare annual salary growth rates between men and women. A basic model examined the unadjusted relationship between gender and salary growth, while a fully adjusted model incorporated explanatory variables such as degree, field, rank, administrative duties, and initial salary. Categorical variables were transformed into dummy variables for numerical analysis. An interaction model tested whether the effect of gender on salary growth varied across professional roles by including interaction terms between gender and rank/field. Additionally, a Generalized Estimating Equations (GEE) model was used to account for repeated salary measurements within individuals over time. To ensure proper estimation, the dataset was reshaped from wide format to long format, converting yearly salary records into individual observations for each faculty member across different years.

4.4 Has sex bias existed in granting promotions from Associate to full Professor?

The dataset was filtered to include only Associate Professors eligible for promotion, excluding Assistant Professors and those promoted after 1990 to ensure sufficient time for promotion. Binary variables were created for sex (Male = 1, Female = 0) and administrative roles (Admin = 1, No Admin = 0), while dummy variables were generated for predictors like academic field and degree type. New variables included years as an Associate Professor and a binary indicator for promotion status, where faculty were considered promoted if they attained Full Professor rank within the dataset timeframe.

Descriptive Analysis:

Descriptive analyses were performed to summarize promotion distributions and compare gender patterns. A histogram showed the distribution of years to Full Professor promotion, while a Kaplan-Meier survival curve analyzed the likelihood of remaining an Associate Professor over time. A bar chart compared promotion rates by gender and a boxplot examined the distribution of years spent as an Associate before promotion.

Inferential Analysis:

A Chi-Square test for independence was conducted to examine if promotion rates differed by gender, specifically comparing the proportion of men and women promoted to Full Professor. The Kaplan-Meier Log-Rank Test was used to analyze gender differences in time-to-promotion, assessing whether one gender experienced significantly longer waiting times. Logistic regression models were built to determine if gender remained a significant predictor of promotion after accounting for additional factors. Four models were tested: Model 1 included only gender, Model

2 added years as an Associate, Model 3 incorporated experience and age at promotion, and Model 4 included all relevant variables like academic field, degree, and administrative roles. Model 2 emerged as the best, showing that gender remained a significant predictor even after adjusting for years spent as an Associate Professor.

5 Results

5.1 Does sex bias exist at the university in the most current year available (1995)?

Descriptive Analysis

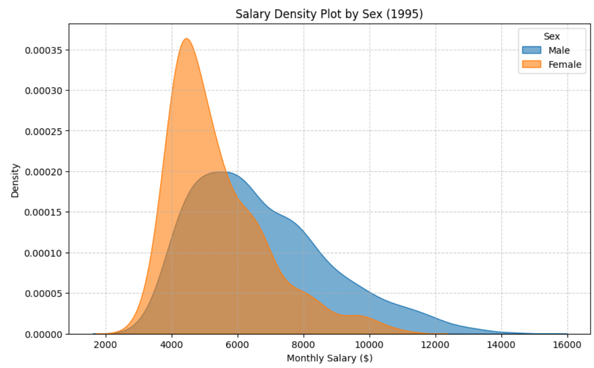
Summary Statistics: Initial descriptive statistics revealed substantial differences in monthly salaries between men and women.

Statistic	Men (n = 1188)	Women (n = 409)
Mean ($)	6732	5397
Median ($)	6313	5016
Standard Deviation ($)	2090	1481

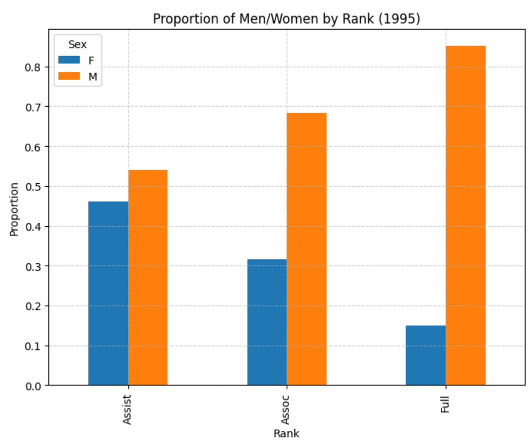
Table 2: Summary statistics of monthly salaries by sex.

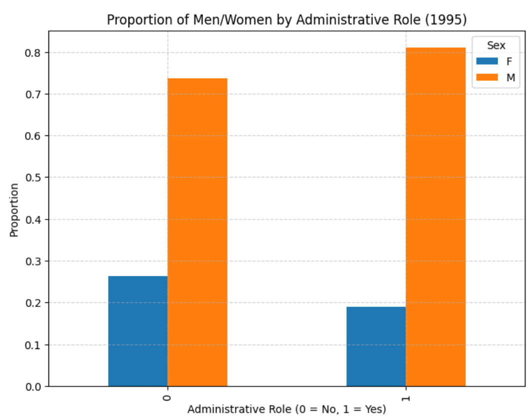
Visualizations:

(a) Salary Distribution by Rank and Sex (1995)



(b) Salary Density Plot by Sex (1995)





Through these visualizations, we can see the following results: Box plots by sex and rank clearly show that men consistently earned more across ranks. Density plots confirm that men had higher salaries overall. Bar charts reveal men disproportionately occupied higher ranks and administrative roles.

Inferential Analysis

Unadjusted Analyses:

Two-sample t-test: A two-sample t-test was conducted to compare the salaries between male and female faculty members at the university in 1995. The test was designed to determine whether there was a statistically significant difference in mean monthly salaries between the two groups. Below are the null and alternative hypotheses:

$$H_0 : \mu_{\text{male}} = \mu_{\text{female}}$$

$$H_1 : \mu_{\text{male}} \neq \mu_{\text{female}}$$

A highly significant difference in mean salaries was observed between men and women ($t = 14.04$, $p < 0.001$ (5.5×10^{-41})).

Simple Linear Regression:

The simple linear regression model used is given by:

$$\text{Salary} = 5396.90 + 1334.73\ (\text{Sex}) + \epsilon$$

The regression result indicated that, on average, men earn approximately $1335 more per month than women, without adjusting for any other factors. This difference is statistically significant ($p < 0.001$ (1.81×10^{-31})). The low R^2 (8.2%) suggested that while sex alone explained some of the salary differences, there were likely other important variables (such as rank, administrative duties, degree type, and field) that also significantly influence salary.

Adjusted Multiple Regression Analysis:

This adjusted model includes relevant variables to determine if the observed salary difference by sex remained significant after controlling for confounders such as Rank (Assistant, Associate, Full), Administrative duties (Yes/No), Degree type (PhD, Professional Degree, Other), Field (Arts, Professional School, Other)

The multiple regression model used is given by:

$$\begin{aligned} \text{Salary} = {} & 3437.03 + 383.52(\text{Sex}) + 571.72(\text{Rank}_{Assoc}) \\ & + 2537.96(\text{Rank}_{Full}) + 1173.13(\text{Admin}) \\ & + 230.11(\text{Degree}_{PhD}) + 840.04(\text{Degree}_{Prof}) \\ & + 705.36(\text{Field}_{Other}) + 1669.30(\text{Field}_{Prof}) \end{aligned} \tag{1}$$

The key hypothesis tested was as follows:

$$H_0 : \beta_1 = 0$$

$$H_A : \beta_1 \neq 0$$

After adjusting for rank, administrative duties, degree type, and academic field, the gender salary gap persisted but reduced significantly. The model explains 52% of the variation in salaries, and the gender salary gap remained statistically significant ($p < 0.001$), with men earning $298 more per month than women on average. Higher ranks (Associate/Full) and administrative duties are associated with large salary increases. Fields like Professional

Schools and "Other" fields pay significantly more than Arts, while holding a PhD or professional degree leads to higher salaries compared to lower degrees. This result highlights persistent gender bias in salaries at the university, even after accounting for other confounding factors.

Interaction Effects

Sex and Administrative Duties: A significant interaction was found ($p = 0.003$). Men holding administrative roles earned an additional premium of $863/month compared to women in similar roles.

1. **Interpretation of Main Effects in the regression model:** The model explains 51.7% of the variation in salary, which is similar to the previous adjusted model without interaction terms. The regression model explains 51.7% of the variation in salary, which is similar to the previous adjusted model without interaction terms. Regarding sex (`sex_binary`), the coefficient is $305.72 with a p-value of 0.001. This indicates that men earn, on average, $305.72 more per month than women, holding other factors constant, including rank, degree, field, and administrative duties. For administrative duties (`admin_1`), the coefficient is $475.60 with a p-value of 0.072. Faculty members with administrative roles earn approximately $475.60 more per month compared to those without administrative duties. However, this effect is marginally significant ($p = 0.072$), indicating weak statistical evidence for this difference.

2. **Interaction Effect:** The interaction term between sex and administrative duties (`sex_admin`) has a coefficient of $863.44 with a p-value of 0.003, which is positive and statistically significant. This suggests that the salary premium associated with administrative duties differs significantly by sex. Specifically, men who hold administrative positions earn an additional $863.44 per month compared to women who hold similar administrative positions, highlighting a notable disparity in how administrative roles contribute to salary differences between men and women in this context.

3. **Calculating the Combined Effect Clearly:** The total salary premium for administrative duties separately by sex is given below:

Sex	**Admin Role Premium Calculation**	**Total Premium**
Female	Admin_1	$475.60
Male	Admin_1 + Sex_Admin	$475.60 + $863.44 = $1,339.04

Table 3: Salary Premium for Administrative Duties by Sex

We tested whether the salary premium associated with administrative duties significantly differs between men and women. The analysis clearly shows that men receive a substantially larger salary premium ($1,339/month) for holding administrative positions compared to women ($476/month).

Sex and Academic Field: Interaction terms were not statistically significant ($p > 0.20$), indicating no strong evidence that the size of the gender gap varied significantly across fields.

1. **Interpretation of Main Effects in the regression model:** The regression model explains 51.7% of the variation in salary, which is consistent with the previous adjusted model without interaction terms.

Regarding the main effects, the variable for sex (`sex_binary`) has a coefficient of \$305.72 with a p-value of 0.001, indicating that men earn, on average, \$305.72 more per month than women, holding other factors constant, such as rank, degree, field, and administrative duties. Administrative duties (`admin_1`) also influence salaries, with a coefficient of \$475.60 and a p-value of 0.072. Faculty members with administrative roles earn approximately \$475.60 more per month compared to those without administrative duties. However, this effect is marginally significant ($p = 0.072$), suggesting weak statistical evidence for this difference. Additionally, the interaction term between sex and administrative duties (`sex_admin`) has a coefficient of \$863.44 with a p-value of 0.003, which is positive and statistically significant. This finding indicates that the salary premium associated with administrative duties differs significantly by sex. Specifically, men who hold administrative positions earn an additional \$863.44 per month compared to women who hold similar administrative positions.

2. **Interaction Effect (Sex and Field):**

Interaction Term	Coefficient (\$/month)	p-value
sex_field_Other	283.71	0.204
sex_field_Prof	246.39	0.426

Table 4: Interaction Effects for Sex and Field Terms in Regression Analysis

Both interaction terms (sex_field_Other and sex_field_Prof) are not statistically significant. This indicates that the gender salary gap does not significantly differ across these academic fields.

We tested whether the size of the gender gap significantly differs between fields (i.e., Is the difference between men's and women's salaries significantly larger in one field compared to another?) and found that while there were gender gaps, the differences in these gaps across fields were not statistically significant. In other words, the analysis didn't find strong evidence that one field had a dramatically larger or smaller gender gap compared to another.

Stratified Analyses

Separate analyses within each rank, administrative role, and field consistently showed significant gender salary gaps:

By Rank: Salary differences by rank reveal notable disparities between men and women. Among Assistant Professors, men earned \$271 more per month than women, with statistical significance ($p = 0.024$). For Associate Professors, the gap was larger, with men earning \$462 more per month compared to women ($p < 0.001$). Full Professors showed the most pronounced disparity, as men earned \$875 more per month than women ($p < 0.001$). These findings indicate that salary differences persist across all ranks, with the gap increasing at higher academic levels.

By Administrative Roles: Salary disparities were also analyzed based on administrative roles. For faculty members with administrative responsibilities, men earned \$1688 more per month compared to women ($p < 0.001$).

The baseline salary for women in administrative roles was \$6769 per month. These results suggest that the gender salary gap is particularly pronounced among faculty with administrative duties, indicating that men disproportionately benefit from these roles. For faculty members without administrative responsibilities, men earned \$1226 more per month than women ($p < 0.001$), with a baseline salary of \$5280 per month for women. Although the gap is smaller in this group compared to those with administrative roles, it remains statistically significant.

By Field: Analysis by academic field revealed significant gender-based salary gaps. In Arts fields, men earned \$578 more per month than women ($p = 0.001$); in Other Fields, the gap was \$1302 ($p < 0.001$); and in Professional Fields, it was \$932 ($p = 0.007$). These findings demonstrate consistent gender bias across all fields. However, interaction analysis showed that while the magnitude of the salary gap varied numerically (\$578 vs \$1302 vs \$932), these differences were not statistically significant across fields, indicating no field is significantly "worse" or "better" in terms of gender bias.

Residual Analysis

The residual analysis revealed that the residuals from the adjusted regression model were not normally distributed. This violation of normality suggests that standard errors may be unreliable under traditional assumptions. To address this, heteroscedasticity-robust standard errors were applied in subsequent models, ensuring more reliable inference despite the non-normality of residuals.

Multicollinearity and VIF Results

The Variance Inflation Factor (VIF) analysis showed moderate multicollinearity among some variables, particularly between the administrative role (`admin_1`) and its interaction term (`sex_admin`), with VIF values around 5.3. However, multicollinearity was not severe enough to invalidate the model, as all other predictors had VIF values below 2. This indicates that the regression model is stable and interpretable, though care should be taken when interpreting coefficients for correlated terms like `admin_1` and `sex_admin`.

Robust Standard Errors

The residual analysis revealed violations of normality and heteroscedasticity in the adjusted regression model. To address these issues, heteroscedasticity-robust standard errors (HC1) were applied. Using robust standard errors ensured that the coefficient estimates remained unbiased, and the p-values reflected more reliable statistical inference despite non-normal residuals. For example, the standard error for the variable `admin_1` decreased from 264.49 to 200.22, and for `sex_admin`, it decreased from 292.99 to 247.55, improving the precision of these estimates while maintaining their statistical significance.

In conclusion, the analysis of faculty salaries at the university in 1995 reveals clear evidence of sex bias. Men consistently earned more than women, with an unadjusted salary gap of \$1335/month and an adjusted gap of \$298/month after controlling for rank, administrative roles, degree type, and academic field. Women were disproportionately represented in lower-paying ranks (Assistant Professors) and fields (Arts), while men dominated higher-paying ranks (Associate and Full Professors) and fields (Professional and Other). Men also disproportionately held administrative roles, receiving a significantly larger salary premium ($1688/month) compared to women in similar positions. These findings indicate structural inequities in salary distribution, representation, and compensation practices at the university.

5.2 Has sex bias existed in the starting salaries of faculty members?

Descriptive Analysis

Summary statistics

The filtered dataset consisted of 1,107 observations with 753 male and 354 female faculty members. The basic summary statistics reveal a salary bias, with male faculty earning an average of $4,015.55 compared to $3,600.32 for female faculty. Male salaries have a higher range and standard deviation, indicating a wider spread of salaries with a maximum salary of $14,464 for males and $8,896 for females.

Parameter	**Male**	**Female**
Count	753	354
Mean	4015.55	3600.32
Standard Deviation	1933.31	1347.82
Minimum	1300	1267
Maximum	14464	8896

Table 5: Summary Statistics of Salaries by Gender

Visualizations Figure 2 scatter plot shows the distribution of starting salaries of male and female faculties over start years. Salaries appear to increase as the start year moves from the late 1970s to the mid-1990s. This is a common pattern, reflecting general salary growth over time (e.g., due to inflation, changes in institutional budgets or evolving market rates). Within each year, there is some spread or range in salaries for both males and females. However, the highest salaries at each time point tend to be dominated by the male group while the lower end overlaps more for both groups.

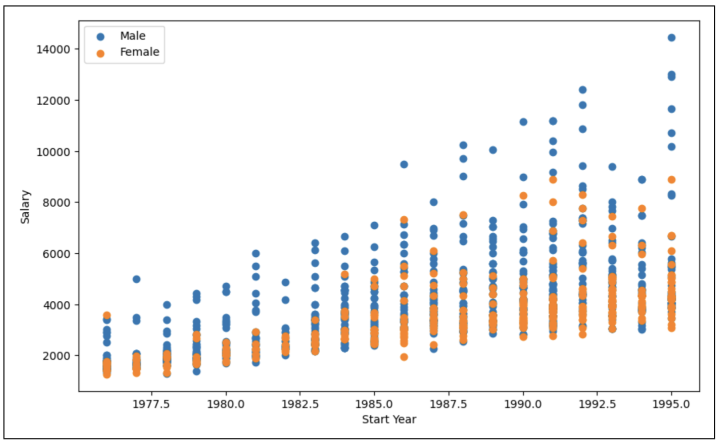
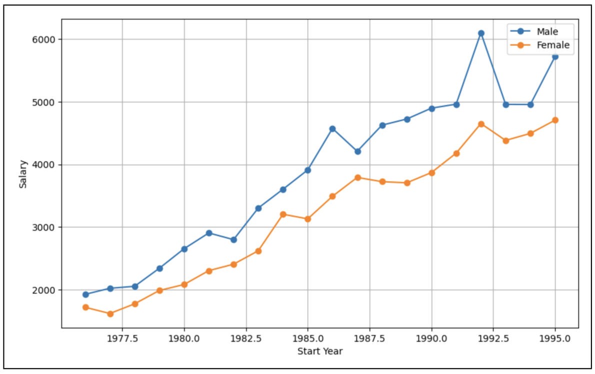


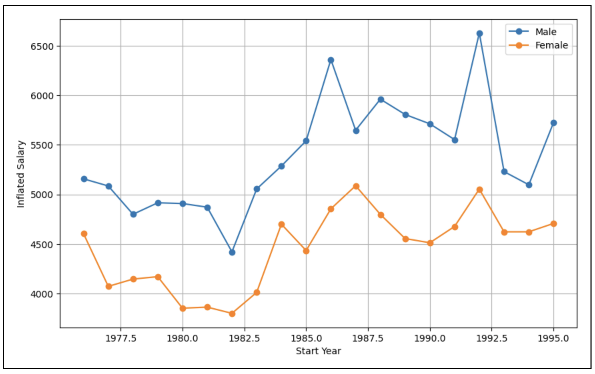
Figure 2: Starting salary distribution over years by gender

Figure 3 provides shows the faculty starting salaries over time, comparing male and female salaries. Figure 3a shows a steady increase for both groups but male salaries remain consistently higher than female salaries. Figure

3b accounts for purchasing power changes over time, revealing that while salaries fluctuate, the gender gap persists. This suggests that male faculty have historically received higher starting salaries, a pattern that continues even after adjusting for inflation.



(a) Starting Salary



(b) Starting Salary after adjusting for inflation

Figure 3: Mean salary distribution over years by gender

Inferential Statistics

Hypothesis Tests

The Welch's t-test Table 6 was conducted to compare the starting salaries between male and female faculty members. The results were assessed both before and after adjusting for inflation. Below is the null and alternate hypothesis.

$$H_0 : \mu_{\text{male}} = \mu_{\text{female}} \ ; \ H_A : \mu_{\text{male}} \neq \mu_{\text{female}}$$

Statistic	**Without Adjusting for Inflation**	**After Adjusting for Inflation**
Sample Mean Difference	415.22	869.09
T-statistic	4.13	9.12
P-value	3.90e-05	4.06e-19
Critical Value	1.96	1.96
Confidence Interval	(218.04, 612.40)	(682.00, 1056.19)
Reject H_0	Yes	Yes

Table 6: Hypothesis Test Results for Starting Salaries by Gender at 95% confidence interval

The results suggest a significant difference in the starting salaries between the two groups. The confidence interval for the mean difference further supports this finding. After adjusting for inflation, the p-value decreased significantly providing even stronger evidence to reject the null hypothesis. The confidence interval also reinforces the conclusion that a significant salary difference exists between male and female faculty members, even when accounting for inflation.

Both analyses clearly indicate that male faculty members, on average, receive higher starting salaries than female faculty members. This reinforces the importance of adjusting for inflation in salary comparisons.

Regression Analysis

A series of regression analyses was conducted using inflation-adjusted salaries as the dependent variable. We began with a simple model then progressively added additional factors and interaction terms to determine whether the male–female salary gap varied under specific conditions or categories.

1. Baseline Model : Salary was regressed on sex as a categorical variable to assess whether there is a significant difference in starting salaries between male and female faculty members.

$$\text{Inflation Adjusted Salary} = 4570.68 + 869.09(\text{Male}) + \epsilon$$

The results showed that the coefficient for the male category was positive (approximately 869), indicating that male faculty members received higher starting salaries on average compared to their female counterparts. The p-value associated with this coefficient was below the conventional significance level (0.05), demonstrating its statistical significance. However, the model's R-squared value was relatively low (0.052), implying that while sex is a significant predictor of salary, other factors, such as experience, rank, and field, also contribute to salary differences.

Multicollinearity and VIF Results

The Variance Inflation Factor (VIF) analysis was conducted (Table 7) to assess multicollinearity among the predictor variables. Initially, the degree (PhD) and start year variables exhibited high VIF values of 10.50 and 14.97, respectively, indicating potential multicollinearity concerns. To mitigate this, the start year variable was centered which significantly reduced its VIF to 1.03, effectively addressing the issue. The VIF for degree (PhD) also decreased from 10.50 to 6.19. Centering the start year variable successfully improved model stability making the regression estimates more reliable.

Feature	VIF (Before Centering)	VIF (After Centering)
sex_M	3.40	3.24
rank_Assoc	1.28	1.28
rank_Full	1.42	1.43
field_Other	6.46	5.20
field_Prof	2.91	2.64
deg_PhD	10.50	6.19
deg_Prof	1.75	1.48
startyr	14.97	1.03
admin	1.26	1.26

Table 7: Variance Inflation Factor (VIF) Results

2. Main-Effects Model (Multiple Covariates) : Next, the regression model is expanded to include additional covariates including field of specialization, academic rank, highest degree earned, administrative duties, years of experience and the start year of the faculty. These variables help account for many of the legitimate reasons salaries might differ across individuals. For example, rank and administrative roles can carry higher pay and more experience can translate into a higher starting salary in some fields. The model used:

$$\begin{aligned}\text{Inflation Adjusted Salary} = {} & 63390 + 227.16(\text{Male}) + 412.64(\text{Field: Other}) + 1049.99(\text{Field: Prof}) \\ & + 932.28(\text{Rank: Assoc}) + 3350.05(\text{Rank: Full}) + 270.42(\text{Degree: PhD}) \\ & + 1034.81(\text{Degree: Prof}) - 601.83(\text{Experience}) + 879.10(\text{Admin}) \\ & + 66.43(\text{Start Year (Centered)}) - 29.87(\text{Year Degree}) + \epsilon\end{aligned}$$

The coefficient for the male category (approximately 227, p=0.002) remained statistically significant. The R^2 value is 0.653 indicating that these additional covariates explained about 65% of the variance in starting salaries. However, despite controlling for these factors, the coefficient for sex (male vs. female) remained statistically significant and continued to indicate higher salaries for male faculty. This persistence implies that the salary gap is not merely a product of differences in qualifications or professional responsibilities. Instead, there appears to be a systematic difference favoring male faculty that is not explained by the covariates included in the model.

3. Interaction Model : After establishing that sex remained a significant predictor even when multiple covariates were included, interaction terms were introduced to determine whether the effect of sex on salary varied under different conditions or categories.

Start Year and Year Degree - The first set of interactions involved sex with degree year and sex with the centered start year. Neither interaction was statistically significant, indicating that the relationship between sex and salary did not differ meaningfully across different degree years or start years.

Experience - After accounting for interaction between sex and experience, the interaction term and the main effect of experience lost statistical significance in that model. Because adding the interaction did not improve the explanatory power of the model and unnecessarily complicated it, this term was dropped for parsimony.

Rank, degree and field - The analysis examined interactions of sex with rank, degree, and field. Only the interaction with Full professors was significant, indicating a pronounced salary gap. The Associate rank showed no significant difference from the Assistant level and interactions with degree and field were insignificant, leading to their exclusion from the final model.

After refining the model to retain only significant interactions, the overall fit explaining roughly 64.9% of the variation in the inflation-adjusted salary. The coefficient for sex remained statistically significant confirming that male faculty members typically have higher starting salaries than their female counterparts. A representative final specification is:

$$\begin{aligned}\text{Inflation Adjusted Salary} = {} & 3705.71 + 168.92 \cdot \text{Sex-Male} + 426.57 \cdot \text{Field-Other} + 1040.25 \cdot \text{Field-Prof} \\ & + 854.77 \cdot \text{Rank-Assoc} + 3054.06 \cdot \text{Rank-Full}) + 148.75 \cdot \text{Degree-PhD} \\ & + 1032.06 \cdot \text{Degree-Prof} + 312.75 \cdot (\text{Sex-Male} \times \text{Rank-Assoc}) + 920.67 \cdot \text{Admin} \\ & + 796.49 \cdot (\text{Sex-Male} \times \text{Rank-Full}) + 41.17 \cdot \text{Start Year Centered} + \epsilon\end{aligned}$$

In conclusion, the analysis strongly suggests a sex bias in faculty starting salaries. Male faculty members consistently earn higher salaries than their female counterparts even after adjusting for inflation and controlling for rank, field, degree, administrative roles and experience. The persistent significance of the sex coefficient in multiple regression models, including those with interactions, indicates that this gap cannot be fully attributed to professional qualifications or responsibilities.

5.3 Has sex bias existed in granting salary increases between 1990 -1995?

Descriptive Analysis

Summary statistics

The dataset included salary data for 1,230 faculty members, of whom 267 were women and 963 were men. Salary slope, defined as the average annual salary increase between 1990 and 1995, was calculated to measure salary progression. On average, men had a slightly higher salary slope ($248.29 per year) compared to women ($232.82 per year). Although this difference of $15.47 per year appears small relative to the variability within each group (standard deviations were $124.46 for women and $129.58 for men), men exhibited a higher maximum salary growth rate compared to women. However, further statistical testing is required to determine whether these differences indicate bias or result from other factors such as rank, field, or administrative roles.

Parameter	**Male**	**Female**
Count	963	267
Mean	248.29	232.82
Standard Deviation	129.58	124.46
Minimum	13.86	56.20
Maximum	1368.8	964.4

Table 8: Summary Statistics of Annual Salary Growth (Salary Slope) by Gender (1990-1995)

Visualizations

noindent Figure 4 provides the the box plot of salary increases from 1990 to 1995 compares salary growth between male and female faculty members. Both groups exhibit similar inter-quartile ranges, indicating comparable salary increase distributions for most individuals. However, men display a slightly higher median salary increase, with a greater spread in the upper range, suggesting more variability. The presence of multiple high-value outliers in both groups indicates that some individuals received significantly larger raises.

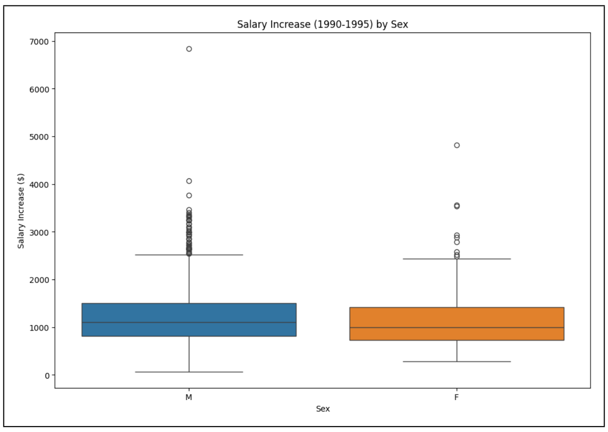


Figure 4: Distribution of salary increases (1990-1995) by gender.

Figure 5 provides a line plot illustrates average salary trends between 1990 and 1995, comparing male and female faculty members. Both genders experienced steady salary growth during this period. However, males consistently maintained higher average salaries across all years. The persistent gap observed between the lines for men and women suggests potential underlying gender-based salary differences, which warrants additional investigation into factors influencing this disparity, such as differences in roles, fields, or career progression opportunities.

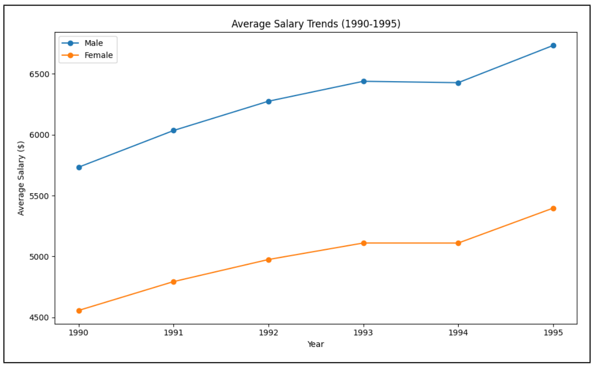


Figure 5: Average salary trends (1990-1995) by gender, showing a persistent gap.

Figure 6 showcases the distributions of annual salary slopes (growth rates) separately for men and women, normalized to account for the different number of observations in each group. Both distributions are right-skewed, indicating most faculty experienced moderate annual salary increases, with fewer cases of exceptionally high growth. The men's distribution peaks at slightly higher salary slopes than women's, suggesting typically higher increases among men. However, substantial overlap exists, reflecting considerable variability within each group.

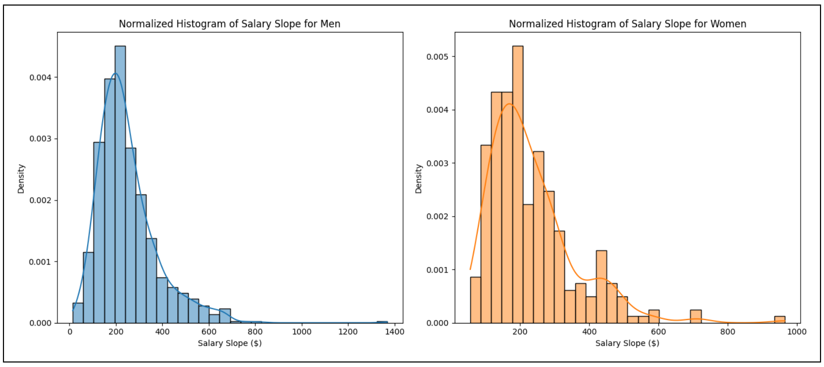


Figure 6: Starting salary distribution over years by gender

Hypothesis Tests:

A two-sample t-test was conducted to determine whether there was a statistically significant difference in the annual salary growth rates between men and women.

$$H_0 : \mu_{\text{male}} = \mu_{\text{female}} \,;\, H_A : \mu_{\text{male}} \neq \mu_{\text{female}}$$

where μ_{male} and μ_{female} represent the mean annual salary growth rates for men and women, respectively. The results of the t-test showed that the mean salary slope for men was \$248.29 per year, while for women, it was \$232.82 per year, with a difference of \$15.47 per year. The t-statistic was 1.7813, and the corresponding p-value was 0.0756. Since the p-value is greater than 0.05, the difference in salary slopes is not statistically significant, meaning that we fail to reject the null hypothesis. This suggests that the observed disparity in salary growth may be due to random variation rather than systematic bias.

Statistic	**Value**
Sample Mean Difference	15.47
Mean Salary Slope (Men)	248.29
Mean Salary Slope (Women)	232.82
T-statistic	1.7813
P-value	0.0756
Critical Value (95%)	1.96
Reject H_0	No

Table 9: Two-Sample T-Test Results for Annual Salary Growth Rate by Gender (1990-1995)

Unadjusted Model Analysis:

To further assess whether gender differences in salary growth exist, an unadjusted regression model was implemented with salary slope as the dependent variable and gender as the sole predictor. The regression equation used in this analysis is:

$$\text{Salary Slope} = \beta_0 + \beta_1 \cdot \text{Sex Dummy} + \epsilon$$

$$\text{Salary Slope} = 248.29 - 15.47(\text{Sex Dummy}) + \epsilon$$

The regression equation modeled the salary slope, defined as the annual salary growth rate. The intercept (β_0) represents the average salary slope for male faculty members. The gender coefficient (β_1) indicates the difference in salary slope between female and male faculty members, with the **Sex Dummy** coded as 1 for females and 0 for males. The error term (ϵ) captures unexplained variation. The results indicate that the mean salary slope for men is $248.29 per year, which corresponds to the intercept of the model. The coefficient for the gender variable is $-15.47, meaning that, on average, women's salary slopes are lower by this amount compared to men. This difference is not statistically significant (p=0.082), as it exceeds the 0.05 threshold. This suggests that, without adjusting for any other factors, the observed difference in salary growth between men and women could be attributed to random variation rather than systematic bias. The model's R-squared value is 0.002, indicating that gender alone explains only 0.2% of the variance in salary growth, reinforcing the conclusion that additional factors may play a more substantial role in determining salary progression.

Adjusted Model (Controlling for Confounders):

The adjusted model incorporates additional explanatory variables, including degree (PhD), field of study, academic rank, administrative role, and initial salary (1990 salary), to account for potential confounding effects on salary growth. This approach allows for a more comprehensive assessment of whether gender differences in salary slopes persist after adjusting for relevant factors.

$$\begin{aligned}\text{Salary Slope} = 88.03 &+ 15.64(\text{Sex Dummy}) + 17.77(\text{PhD}) - 14.44(\text{Arts Field}) \\ &+ 22.42(\text{Professional Field}) + 100.46(\text{Full Professor}) + 48.42(\text{Associate Professor}) \\ &+ 135.12(\text{Admin}) + 0.007(\text{Salary 1990}) + \epsilon\end{aligned}$$

The intercept ($\beta_0 = 88.03$) represents the expected annual salary growth for a male faculty member without a PhD, not in an arts or professional field, not in an administrative role, and with an initial salary of zero (for interpretative purposes). The coefficient for the gender variable ($\beta_1 = 15.64$, $p = 0.055$) suggests that women had a slightly higher salary growth rate than men after adjusting for confounders. However, this difference is not statistically significant at the 0.05 level. The R-squared value of 0.250 indicates that 25% of the variance in salary slopes is explained by these factors, suggesting that rank, administrative roles, and field of study have a more substantial influence on salary growth than gender alone. After controlling for relevant factors, gender is not a significant predictor of salary growth.

Interaction Model Analysis:

To further explore whether gender disparities in salary growth vary across different faculty ranks and academic fields, an interaction model was estimated. This model builds upon the adjusted model by incorporating interaction terms between gender and key categorical variables such as faculty rank and academic discipline. The inclusion of these terms allows for an examination of whether salary growth differences between men and women are more pronounced within specific subgroups.

$$\begin{aligned}\text{Salary Slope} = {} & 99.66 - 13.38(\text{Sex Dummy}) + 16.62(\text{PhD}) - 26.63(\text{Arts Field}) \\ & + 24.83(\text{Professional Field}) + 87.39(\text{Full Professor}) + 42.14(\text{Associate Professor}) \\ & + 135.00(\text{Admin}) + 0.0073(\text{Salary 1990}) + 37.46(\text{Sex} \times \text{Full Professor}) \\ & + 14.26(\text{Sex} \times \text{Associate Professor}) + 35.76(\text{Sex} \times \text{Arts Field}) \\ & - 25.99(\text{Sex} \times \text{Professional Field}) + \epsilon\end{aligned}$$

The results indicate that the baseline salary growth, represented by the intercept ($\beta_0 = 99.66$), corresponds to the expected annual salary increase for a male faculty member without a PhD, not in an arts or professional field, not holding an administrative position, and with an initial salary of zero. The gender coefficient ($\beta_1 = -13.38$, $p = 0.664$) suggests women had slightly lower salary slopes than men, but this difference was not statistically significant. None of the interaction terms showed statistically significant effects, indicating no clear variation in gender-based salary differences across ranks or disciplines. The R-squared value (0.255) indicates the model explains approximately 25.5% of the variance in salary growth, slightly improving upon the adjusted model. However, the absence of significant interaction effects reinforces that gender does not differentially impact salary progression across academic ranks or fields.

Generalized Estimating Equations (GEE) Model:

The Generalized Estimating Equations (GEE) model was used to study salary growth while accounting for repeated salary observations within faculty members. Unlike ordinary least squares (OLS), which assumes observations are independent, the GEE model considers that salaries for the same individual are correlated over time. This approach provides more reliable estimates when analyzing salary trends over multiple years.

$$\begin{aligned}\text{Salary} = {} & 212.28 - 63.57(\text{Sex Dummy}) + 43.19(\text{PhD}) - 51.42(\text{Arts Field}) \\ & + 69.70(\text{Professional Field}) + 283.23(\text{Full Professor}) + 162.00(\text{Associate Professor}) \\ & + 331.99(\text{Admin}) + 1.045(\text{Salary 1990}) + 121.04(\text{Sex} \times \text{Full Professor}) \\ & + 71.01(\text{Sex} \times \text{Associate Professor}) + 83.51(\text{Sex} \times \text{Arts Field}) \\ & - 95.33(\text{Sex} \times \text{Professional Field}) + \epsilon\end{aligned}$$

The model indicates that initial salary, rank, and administrative roles strongly influenced faculty salary growth. Gender alone did not significantly impact salary, although female full professors earned relatively higher salaries than their male peers. Specialization in professional fields and holding a PhD slightly increased salaries, though these effects weren't statistically significant. Faculty in administrative roles saw the highest salary growth, emphasizing the financial benefits of leadership positions. The strongest predictor was the starting salary in 1990, highlighting the lasting impact of early-career earnings. These results underline the role of structural factors, such as rank and administrative responsibilities, in shaping salary trends over time.

Overall, the analysis showed no statistically significant gender differences in salary growth after adjusting for factors such as degree, field, administrative role, and initial salary. While men had slightly higher salary slopes on average, the difference was not significant. Interaction analyses found no consistent gender differences across academic fields or ranks, though full professors who were women had relatively higher salary growth. The GEE model, accounting for repeated measures, reinforced that salary progression was mainly influenced by structural factors like rank, administrative roles, and initial salary rather than gender alone. Overall, the findings suggest that observed differences in salary growth were largely driven by career-related variables rather than systemic gender bias.

5.4 Does sex bias affect the likelihood of promotion from Associate Professor to Full Professor?

Descriptive Analysis:

Summary statistics:

The filtered dataset consisted of 5064 male and 1465 female faculty members. The summary statistics reveal a notable disparity in promotion rates, with male faculty achieving a higher promotion rate compared to their female counterparts. The average time spent as an Associate Professor before promotion was higher for men (10.136 years) compared to women (9.192 years). This suggests that men tend to remain in the Associate rank for a longer period before achieving promotion. The standard deviation of promotion time suggests a wider spread for women, reflecting greater variability in the time taken for promotion.

Parameter	**Male**	**Female**
Count	5064	1465
Mean years as Associate	10.136	9.192
Standard Deviation	6.524	5.394
Minimum	0	0
Maximum	45	35
Promotion Rate %	48.69	42.1

Table 10: Summary Statistics of Annual Salary Growth (Salary Slope) by Gender (1990-1995)

Visualizations:

Figure 4 presents a histogram of the number of years faculty members spent as Associate Professors before promotion, categorized by gender. The distribution is right-skewed, indicating that most promotions occur within the first 10 to 15 years, but some faculty remain at the Associate rank for over 30 years. The density curves reveal that male faculty tend to have a broader distribution, with more cases at higher years before promotion, while female faculty appear to cluster in the lower-to-mid range of promotion years. This suggests potential differences in the career progression timeline between genders.

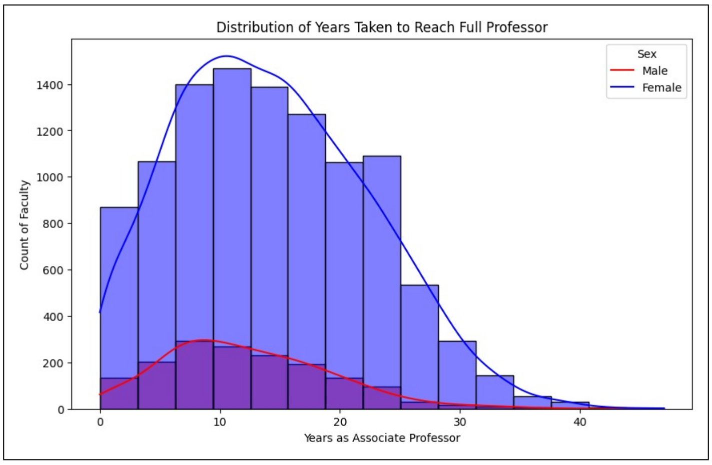


Figure 7: Histogram of years taken to reach full professor.

Figure 8 presents the Kaplan-Meier survival analysis, which estimates the probability of remaining an Associate Professor over time. The curve shows that male faculty (blue) tend to remain in the Associate rank for longer periods than female faculty (orange). The steeper decline for female faculty suggests greater variability in career progression, which could result from either faster promotions or higher attrition rates from academia. The long tail in both distributions indicates that while most faculty achieve promotion within the first 10 to 15 years, a subset remains in the Associate rank for significantly longer durations. Further analysis is needed to disentangle whether the observed gender differences stem primarily from promotion dynamics or faculty turnover.

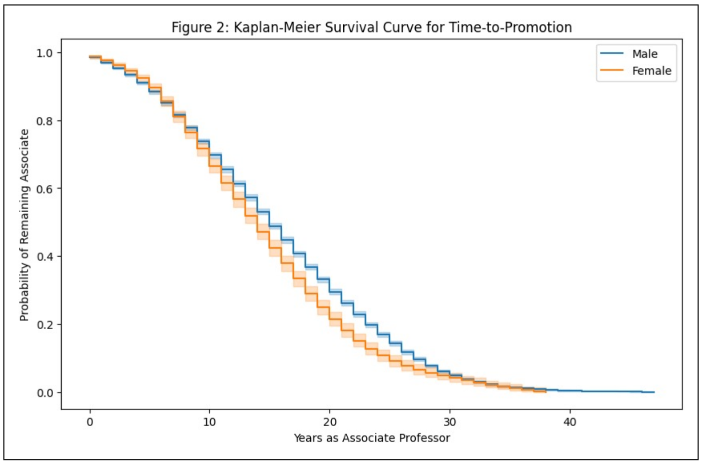


Figure 8: Kaplen Meier Survival Curve for Time-to-Promotion

Figure 9 presents a bar chart comparing the overall promotion rates for male and female faculty. Male faculty exhibit a higher promotion rate (approximately 80%) compared to female faculty (around 65%). The difference suggests a potential gender disparity in promotion likelihood, which may be further explored through statistical modeling to determine whether this gap persists after adjusting for factors such as experience, academic field, and administrative roles.

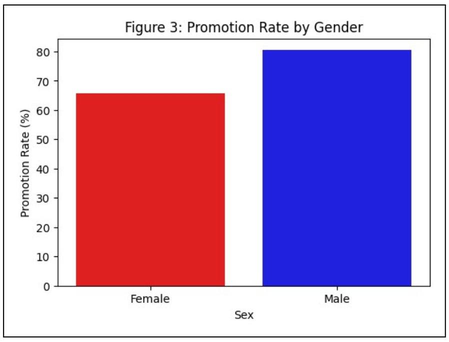


Figure 9: Distribution of promotion time by gender

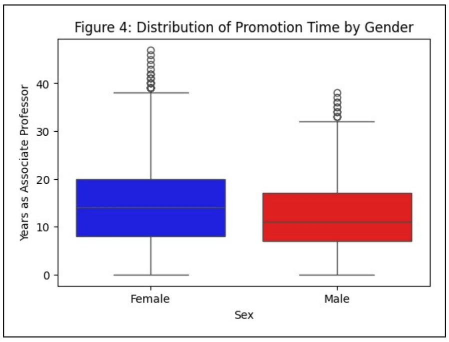


Figure 10: Promotion rate by gender

Figure 10 illustrates the distribution of the number of years faculty spent as Associate Professors before promotion, using box plots for each gender. The median promotion time is lower for female faculty than for male faculty, indicating that women, on average, may achieve promotion slightly sooner. However, the presence of outliers in both distributions suggests that some faculty members experience significantly longer delays in promotion. The interquartile range (IQR) for men is wider, reflecting a greater spread in promotion times compared to women.

Hypothesis Tests:

A two-sample t-test was conducted to compare the time taken for promotion from Associate Professor to Full Professor between male and female faculty members. The null and alternative hypotheses were:

$$H_0 : \mu_{\text{male}} = \mu_{\text{female}} \,; H_A : \mu_{\text{male}} \neq \mu_{\text{female}}$$

The test results revealed a highly significant difference in promotion times between men and women ($t = 10.91, p < 0.001$) , confirming that their career trajectories differ. However, further modeling is required to determine whether this gap persists after controlling for experience, academic field, and administrative roles. A Chi-Square test was performed to examine whether promotion rates significantly differ by gender. The null and alternative hypotheses were:

$$H_0 : P_{\text{male}} = P_{\text{female}} \,; H_A : P_{\text{male}} \neq P_{\text{female}}$$

The test yielded a statistically significant result ($\chi^2 = 266.08, p < 0.001$), indicating that men and women do not experience the same likelihood of promotion.

Simple Logistic Regression: Effect of Gender on Promotion:

A simple logistic regression was performed to assess the effect of gender on the likelihood of promotion. The model was specified as:

$$\text{Promotion} = 0.646 + 0.767(\text{Sex})$$

The results showed that gender is a significant predictor of promotion. The odds of promotion were significantly higher for male faculty compared to female faculty ($\beta_1 = 0.7672, p < 0.001$).

Adjusted Multiple Logistic Regression Analysis:

To account for additional factors influencing promotion, a multiple logistic regression model was estimated:

$$\text{Promotion} = -0.3716 + 0.6438(\text{Sex}) + 0.0194(\text{Years as Associate}) + 0.0459(\text{Age at Promotion}) + 0.0264(\text{Experience})$$

The results indicated that even after controlling for years spent as an associate professor, age at promotion, and experience, gender remained a significant predictor of promotion ($\beta_1 = 0.6438, p < 0.001$). A full model was then developed, including additional factors such as academic field, degree type, and administrative duties:

$$\begin{aligned}\text{Promotion} = &-1.27 + 0.5195(\text{Sex}) + 0.0216(\text{Years as Associate}) + 0.0481(\text{Age at Promotion}) \\ &+ 0.0265(\text{Experience}) + 0.4724(\text{Field}) + 0.4295(\text{Degree}) + 0.4226(\text{Admin Role})\end{aligned}$$

In this full model, the effect of gender was slightly reduced but remained statistically significant ($\beta_1 = 0.5195, p < 0.001$), suggesting that even after controlling for experience and qualifications, men continue to have a higher likelihood of promotion than women.

Interaction Effects:

To determine whether the gender disparity in promotions varied by other factors, we tested interaction effects between gender and administrative duties, as well as gender and academic field.

Sex and Administrative Duties:

An interaction term was introduced between sex and administrative duties to assess whether the gender gap in promotion rates differs between faculty with and without administrative roles. The interaction model is:

$$\text{Promotion} = 0.6 + 0.7441(\text{Sex}) + 0.412(\text{Admin Role}) + 0.205(\text{Sex } and \text{ Admin Role})$$

The results showed: The main effect of sex remained significant ($\beta_1 = 0.7441, p < 0.001$), showing that men were still more likely to be promoted than women. The main effect of administrative duties was also significant ($\beta_2 = 0.412, p < 0.001$), indicating that faculty in administrative positions had a higher likelihood of promotion. The interaction term was statistically significant ($\beta_3 = 0.205$), indicating that administrative roles increase the likelihood of promotion more for men than for women.

Sex and Academic Field:

Another interaction model tested whether the effect of gender on promotion varied across different academic fields. The interaction model is:

$$\text{Promotion} = 0.83 - 0.0763(\text{Sex}) - 0.2012(\text{Field}) + 0.8460(\text{Sex } and \text{ Field})$$

The analysis revealed significant gender disparities in faculty promotions. Women take longer to be promoted from Associate to Full Professor than men, a difference confirmed by descriptive statistics and logistic regression models. Even after controlling for experience, administrative roles, academic field, and degree type, men remain significantly more likely to be promoted. Interaction analyses showed that administrative roles provide a larger career advantage to men than women, amplifying the gender gap. While gender disparities exist across all academic fields, the effect does not significantly vary between disciplines ($p > 0.05$). Structural and institutional factors likely contribute to these disparities, highlighting the need for policy interventions to ensure equitable career advancement.

6 Conclusion

In conclusion, the analysis revealed sex bias in 1995 faculty salaries, where men earned significantly more than women, even after adjusting for rank, administrative roles, and academic field, pointing to structural inequities in pay distribution and representation. Starting salaries also exhibited a persistent gender gap, with men earning higher salaries than women despite controlling for professional qualifications and experience, suggesting systemic disparities at the point of hiring. However, salary growth between 1990 and 1995 did not show statistically significant gender differences after adjusting for factors like degree, field, administrative roles, and initial salary. While men had slightly higher salary slopes on average, this difference was not significant, and rank and administrative roles were the main drivers of salary progression. In contrast, promotion trends showed significant disparities, with women taking longer to advance from Associate to Full Professor, even after accounting for experience and other career-related factors. Additionally, administrative roles provided a larger career advantage to men, further widening the gender gap in advancement. Overall, while salary increases over time were largely shaped by structural factors rather than gender alone, the broader patterns in hiring, salary distribution, and promotions indicated persistent institutional barriers for women in academia.

Limitations of the study include potential unobserved factors influencing salary growth and promotion, such as performance evaluations, grant funding, or negotiation strategies, which were not accounted for in the models. Additionally, while the study controlled for major structural variables, it is possible that discipline-specific or institutional policies further contributed to gender disparities in ways not fully captured. Despite these limitations, the findings highlight systemic gender inequities in hiring, salary distribution, and promotions, underscoring the need for policy interventions to address these barriers.

7 Tools Used

For this analysis several tools were utilized to facilitate data processing, visualization and model development. Python along with Jupyter Notebook in the Anaconda environment was employed for data manipulation, exploratory analysis and statistical modeling.